Nathaniel Rayman

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Education

University of Waterloo — Waterloo, ON
Bachelor of Applied Science in Mechatronics Engineering — *September 2026 – June 2031*
Relevant Coursework: Engineering Graphics (SolidWorks, AutoCAD)

Skills

Programming Languages: Java, Python, C++, C#
CAD & Design: Fusion360, Onshape, SolidWorks
Tools & Workflow: Git, VS Code, Visual Studio
Embedded: Arduino, Serial/I2C, CAN bus, Soldering

Experience

President & Programming Lead — September 2024 – June 2026
Unionville High School VEX Robotics Team — *Markham, ON*

- Managed 40 general and executive members, alongside managing the code-base with Git and GitHub, allowing collaboration across a team of 8 programmers
- Competed at the provincial level, being one of the top 80 teams in Ontario 3 years in a row
- Designed in Fusion360, programmed in C++, and utilized concepts like PID, pure pursuit, and wheel odometry to localize robots and create autonomous routines
- Updated website with HTML, CSS, and GitHub Pages

General Team Member — September 2025 – June 2026
Team Minekee First Robotics Competition Team — *North York, ON*

- Used Java with WPILib to assist with robot programming
- Tuned LimeLight camera configurations on the robot, allowing the robot to see AprilTags to fuse with the IMU data and drivebase data to localize itself on the field
- Ranked in the top 22% of teams in Ontario, allowing us to compete internationally
- Scouted matches and strategized with team to ensure matches went in our favour
- Helped prototype, assemble, and wire up intake subsystem on the robot using the CAN bus protocol

Camp Counselor — June 2024 – August 2026
City of Richmond Hill — *Richmond Hill, ON*

- Worked with children ages 4 - 12 at various Richmond Hill community centers
- Communicated with children effectively to ensure they follow instructions, and resolved conflicts when needed
- Entrusted with health and safety responsibilities, including administering first aid and belaying for a ropes challenge course

Projects

Wearable V1 Wings Replica | *Fusion360* — August 2026

- Created mechanical articulating wings based off the video-game character V1 in one week for FanExpo Canada
- Manufactured 3D printed PLA parts, including a hinged backpack, pulleys, and 8 wings
- Designed with repairability in mind, using standard M4 screws and heat-set inserts to allow quick repairs
- Took into consideration print orientation and infill when printing, optimizing parts for strength and weight
- Calculated belt sizing and pulley ratios to create a 1:3 ratio of rotation between wings, and used CAD to visualize wing movement before manufacturing

Fishy Business | *Unity, C#* — November 2024

- Created for the Counterspell Game Jam in 48 hours, winning second place out of 35 other games
- Used ScriptableObjects to allow us to quickly create unique weapons for the player to use
- Utilized Unity Version Control, allowing collaboration and solving merge conflicts

H.A.I.L Glove | *Arduino, C++, C#* — September 2023

- Created a glove that when rotated controls the mouse and keyboard for Hack the North in 48 hours
- Used an Arduino Uno, a MPU6050 gyroscope, and a sewed glove to detect glove rotation
- Wrote Arduino firmware with C++, used C# to create a GUI, and connected the GUI with the Arduino over USB